



Deep-Regulatory: Deciphering Cell-Specific Transcription Factor Binding & Chromatin Accessibility



Research Overview

Gene regulation ka sab se bara challenge ye samajhna hai ke aik hi DNA sequence mukhtalif cell types (e.g., Neurons vs. Cardiomyocytes) mein alag behavior kyun dikhata hai. Yeh project aik hybrid **Deep Learning** framework pesh karta hai jo DNA sequences aur epigenetic signatures ko analyze karke regulatory "hotspots" ko predict karta hai. Iska maqsad non-coding mutations (SNPs) ke impact ko functional biology ke tanazir mein decode karna hai.



The Biological Problem

- **Context-Specific Regulation:** 98% human genome non-coding hai, jahan enhancers aur promoters gene expression ko control karte hain. Inki activity cell-specific hoti hai aur inhein identify karna mehanga aur waqt-talab kaam hai.
- **Variant Interpretation:** Single Nucleotide Polymorphisms (SNPs) aksar regulatory regions mein hote hain, lekin inka effect puray gene regulatory network par kya hota hai, ye predict karna aik bari mushkil hai.



The Solution: A Hybrid Pipeline

Humne aik multi-stage pipeline design ki hai jo computational predictions ko experimental validation se jorti hai:

1. In Silico Modeling (Dry-Lab)

- **Transformer Architecture:** Humne aik sophisticated model banaya hai jo DNA sequences (1000bp windows) aur epigenetic marks (H3K27ac/DNA Methylation) ko read karta hai.
- **XAI Integration: Integrated Gradients** aur **Grad-CAM** ke zariye AI ke decisions ko map kiya gaya hai taake specific TF-binding motifs ko pehchana ja sake.
- **In Silico Mutagenesis:** Har base pair ki aehmiyat janne ke liye automated SNP impact analysis.

2. In Vitro Simulation (Wet-Lab Bridge)

AI ki predictions ko verify karne ke liye humne virtual validation modules design kiye hain:

- **Luciferase Reporter Assays:** Predicted enhancers ki strength ko measure karna.
- **CRISPRi/RT-qPCR:** Specific regulatory elements ko silence karke target gene expression mein kami ko quantify karna.



Dataset & Features

- **Data Source:** High-fidelity synthetic genomic data jo ENCODE aur NCBI GEO datasets ki complexity ko simulate karta hai.
- **Features:** * One-hot encoded DNA sequences.
 - Cell-specific epigenetic signal tracks.
 - Ground-truth Transcription Factor (TF) binding labels.



Tech Stack

- **Deep Learning:** TensorFlow / Keras (Functional API), PyTorch.
- **Genomics Tools:** BioPython, Logomaker (for sequence logos), Scikit-learn.
- **Data Science:** NumPy, Pandas, Matplotlib, Seaborn.
- **Explainable AI:** Custom GradientTape implementations for saliency mapping.



Key Research Features

- **Multi-Modal Integration:** DNA sequence aur epigenetic signals ka combined analysis.
- **High-Throughput VEP:** Variant Effect Prediction jo SNPs ke pathogenicity score calculate karta hai.
- **Visual Research Dashboard:** Sequence logos aur virtual Western Blot/Gel simulations jo clinical presentation ke liye taiyar hain.



Future Research & Impact

- **Precision Medicine:** Cancer genetics aur neuro-developmental disorders mein personalized regulatory maps banana.
- **Drug Discovery:** Aise ligands design karna jo specifically mutated regulatory regions ko target kar saken.
- **Live Validation:** Is pipeline ko real-world **ChIP-seq** aur **ATAC-seq** data par deploy karna.